Data Understanding

Summary Statistics

There are 334 observations in the dataset, with 18 features, split into 8 categorical features, and 10 numerical features. Shapes and summary statistics below:

```
(334, 18)
```

Categorical Summary Statistics

	Status	Drug	Sex	Ascites	Hepatomegaly	Spiders	Edema	Stage
count	334	254	334	254	254	254	334	329.0
unique	2	2	2	2	2	2	3	4.0
top	1	D-penicillamine	F	N	N	N	N	3.0
freq	205	131	300	236	128	180	284	126.0

Numerical Summary Statistics

	Age	Bilirubin	Cholesterol	Albumin	Copper	Alk_Phos	SGOT	Triglycerides	Platelets	Prothrombin
count	334.000000	334.000000	232.000000	334.000000	252.000000	254.000000	254.000000	232.000000	325.000000	332.000000
mean	18458.628743	3.353892	373.297414	3.506437	98.162698	1967.518898	123.388425	124.668103	260.218462	10.764458
std	3860.211171	4.551418	247.550620	0.423248	89.603383	2130.869240	57.066844	67.516406	98.882826	1.080674
min	9598.000000	0.300000	120.000000	1.960000	9.000000	289.000000	26.350000	33.000000	62.000000	9.000000
25%	15593.500000	0.800000	247.000000	3.260000	41.500000	862.500000	80.485000	84.000000	192.000000	10.000000
50%	18414.000000	1.350000	308.500000	3.550000	71.500000	1288.500000	116.250000	108.000000	252.000000	10.600000
75%	21231.500000	3.575000	396.750000	3.777500	121.000000	2031.500000	153.450000	149.000000	319.000000	11.100000
max	28650.000000	28.000000	1775.000000	4.640000	588.000000	13862.400000	457.250000	598.000000	721.000000	18.000000

All Pearson correlations against target variable Status:

```
Correlation between Status and Age: -0.2809514234267491, P-value: 1.735581049070844e-05
Correlation between Status and Bilirubin: -0.4337976285361258, P-value: 7.841160149687726e-12
Correlation between Status and Cholesterol: -0.1764885116614191, P-value: 0.007690994148159873
Correlation between Status and Albumin: 0.2739025089973419, P-value: 2.860795935613433e-05
Correlation between Status and Copper: -0.3277730389584504, P-value: 4.38638256316375e-07
Correlation between Status and Alk_Phos: -0.2798756890296976, P-value: 1.874784573579679e-05
Correlation between Status and SGOT: -0.23208996916490288, P-value: 0.000422287091709767
Correlation between Status and Triglycerides: -0.2033012741929929, P-value: 0.0020815389491478753
Correlation between Status and Platelets: 0.0787354795224943, P-value: 0.23738197315813875
Correlation between Status and Prothrombin: -0.4308143990948541, P-value: 1.1278079195152416e-11
```

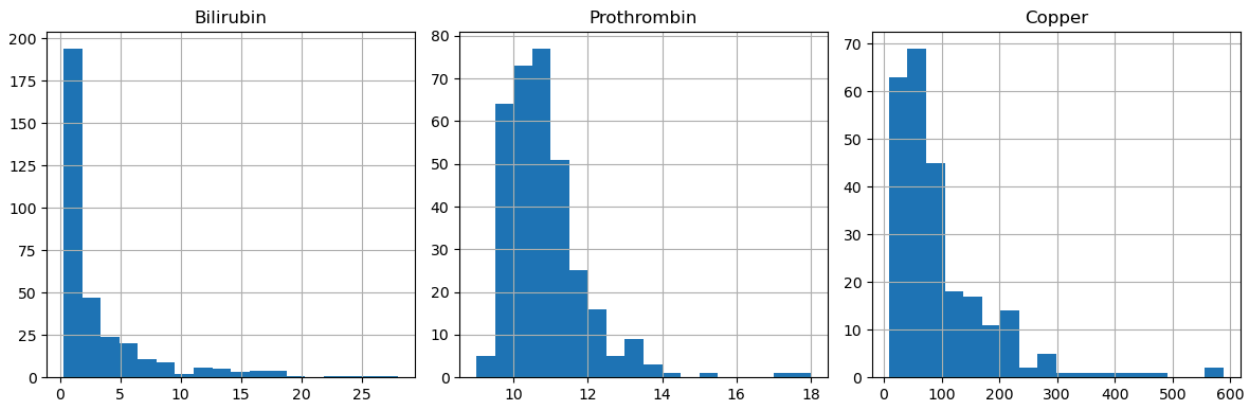
Which gives us our three strongest correlations (found and ordered by absolute value)

```
          Name  Correlation
1    Bilirubin    -0.433798
9  Prothrombin    -0.430814
4       Copper    -0.327773
```

To plot our histograms, first we work out how many bins to use, with the rule of thumb bins = sqrt(n):

```
n = 334
Number of bins: 18
```

Which we use to plot our histograms to observe the shape of the distributions.



- Is data normally distributed?
No, not really. All three distributions have a heavy right skew, severe clustering around 0 and extreme outliers. The distribution of Prothrombin could be approximated as normal, especially compared to the other distributions but it still has the same red flags present. Further EDA should look at
1. Residuals vs. Fitted Plot (Variance & Linearity)
2. Normal Q-Q Plot (Normality)
3. Scale-Location Plot (Variance)
4. Residuals vs. Leverage Plot (Outliers & Influence)
to test normal distribution assumptions.
- Is it unimodal/bimodal/multimodal?
All distributions are unimodal.
- Skewness
All distributions have a positive skewness value much greater than 1, indicating extreme asymmetry with a long tail on the right, with the majority of datapoints clustering on the left of the median.
- Kurtosis
All distributions have a positive kurtosis value much greater than 0, which suggests that they have very heavy tails, with greater concentrations of extreme values/outliers.

Skewness and Kurtosis Values

```
Bilirubin - Skewness: 2.6042, Kurtosis: 7.4481
Prothrombin - Skewness: 2.2818, Kurtosis: 9.7091
Copper - Skewness: 2.3966, Kurtosis: 7.7066
```

To look at missing values in the dataset, we can first use isna().sum() to check the total number of missing values for each feature. A number of features are missing exactly 80 values, with a

couple both missing exactly 102. This would suggest a large number of observations with missing variables have the same values missing across features.

We can then look at the percentage of missing values (against target variable), which would confirm what we have found above - the equal percentages of missing data for $Status_0$ and $Status_1$ show that these missing values are shared across selected observations.

```
df.isna().sum()
✓ 0.0s

Status              0
Drug               80
Age                 0
Sex                 0
Ascites            80
Hepatomegaly       80
Spiders            80
Edema               0
Bilirubin           0
Cholesterol       102
Albumin             0
Copper             82
Alk_Phos           80
SGOT               80
Triglycerides     102
Platelets           9
Prothrombin         2
Stage               5
```

```
Status
0    0.240310
1    0.239024
Name: Drug, dtype: float64
Status
0    0.240310
1    0.239024
Name: Ascites, dtype: float64
Status
0    0.240310
1    0.239024
Name: Hepatomegaly, dtype: float64
Status
0    0.240310
1    0.239024
Name: Spiders, dtype: float64
Status
0    0.302326
1    0.307317
Name: Cholesterol, dtype: float64
Status
0    0.248062
1    0.243902
Name: Copper, dtype: float64
```

```
Status
0    0.240310
1    0.239024
Name: Alk_Phos, dtype: float64
Status
0    0.240310
1    0.239024
Name: SGOT, dtype: float64
Status
0    0.302326
1    0.307317
Name: Triglycerides, dtype: float64
Status
0    0.038760
1    0.019512
Name: Platelets, dtype: float64
Status
0    0.007752
1    0.004878
Name: Prothrombin, dtype: float64
```

What the missing values look like in the dataframe:

	Status	Drug	Age	Sex	Ascites	Hepatomegaly	Spiders	Edema	Bilirubin	Cholesterol	Albumin	Copper	Alk_Phos	SGOT	Triglycerides	Platelets	Prothrombin	Stage
17	0	NaN	16071	M	NaN	NaN	NaN	N	9.5	NaN	3.63	NaN	NaN	NaN	NaN	292.0	10.2	3.0
18	0	D-penicillamine	19577	F	Y	Y	Y	N	1.2	269.0	3.12	NaN	1441.0	165.85	68.0	166.0	11.1	4.0
23	0	Placebo	15857	F	N	Y	Y	S	8.5	NaN	3.34	161.0	1428.0	181.35	NaN	88.0	13.3	4.0
25	1	NaN	15341	F	NaN	NaN	NaN	N	8.1	NaN	2.82	NaN	NaN	NaN	NaN	193.0	10.4	2.0
30	1	NaN	23011	F	NaN	NaN	NaN	N	0.9	NaN	3.58	NaN	NaN	NaN	NaN	224.0	10.0	3.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
324	1	NaN	18993	F	NaN	NaN	NaN	N	1.7	NaN	3.24	NaN	NaN	NaN	NaN	231.0	10.5	3.0
327	0	NaN	23741	F	NaN	NaN	NaN	S	1.4	NaN	3.04	NaN	NaN	NaN	NaN	331.0	12.1	4.0
329	1	NaN	23011	F	NaN	NaN	NaN	N	0.6	NaN	3.94	NaN	NaN	NaN	NaN	234.0	11.5	2.0
330	1	NaN	14975	M	NaN	NaN	NaN	N	1.7	NaN	3.66	NaN	NaN	NaN	NaN	92.0	11.0	4.0
332	0	NaN	21915	F	NaN	NaN	NaN	S	1.8	NaN	3.62	NaN	NaN	NaN	NaN	225.0	9.9	2.0

107 rows × 18 columns

Because the percentage of these missing values is much greater than 5% (20-25%) we will impute the values, rather than remove the observations. I considered removing the observations with only a few missing values, but this had a knock-on negative effect on accuracy across the rest of the analysis and opted to keep them.

Data Preprocessing

Ultimately choosing not to remove any observations from the dataset, we move on to look at potentially removing features from the dataset. Using crosstab to check for any strong correlations between categorical features, we do not find any strong correlations between features that would warrant removing them. I did notice here that there are a much higher number of female patients than male, which may have an effect on feature distributions, but ultimately decided to keep Sex included in the feature set, as removing it had a negative effect on overall accuracy.

I chose to encode all categorical variables as nominal variables. Stage could have been encoded as ordinal to show the progression of the disease but easier for this case to encode as all nominal, and didn't have a significant effect on the model.

I used linear scaling on the data, after checking the distribution of all numerical variables with histograms, as well as the top three correlated variables discussed above, we have some suggestion of normally distributed data, but the majority of features violate assumptions of normal distribution, so it would be inappropriate to use min max scaling.

Feature Ranking

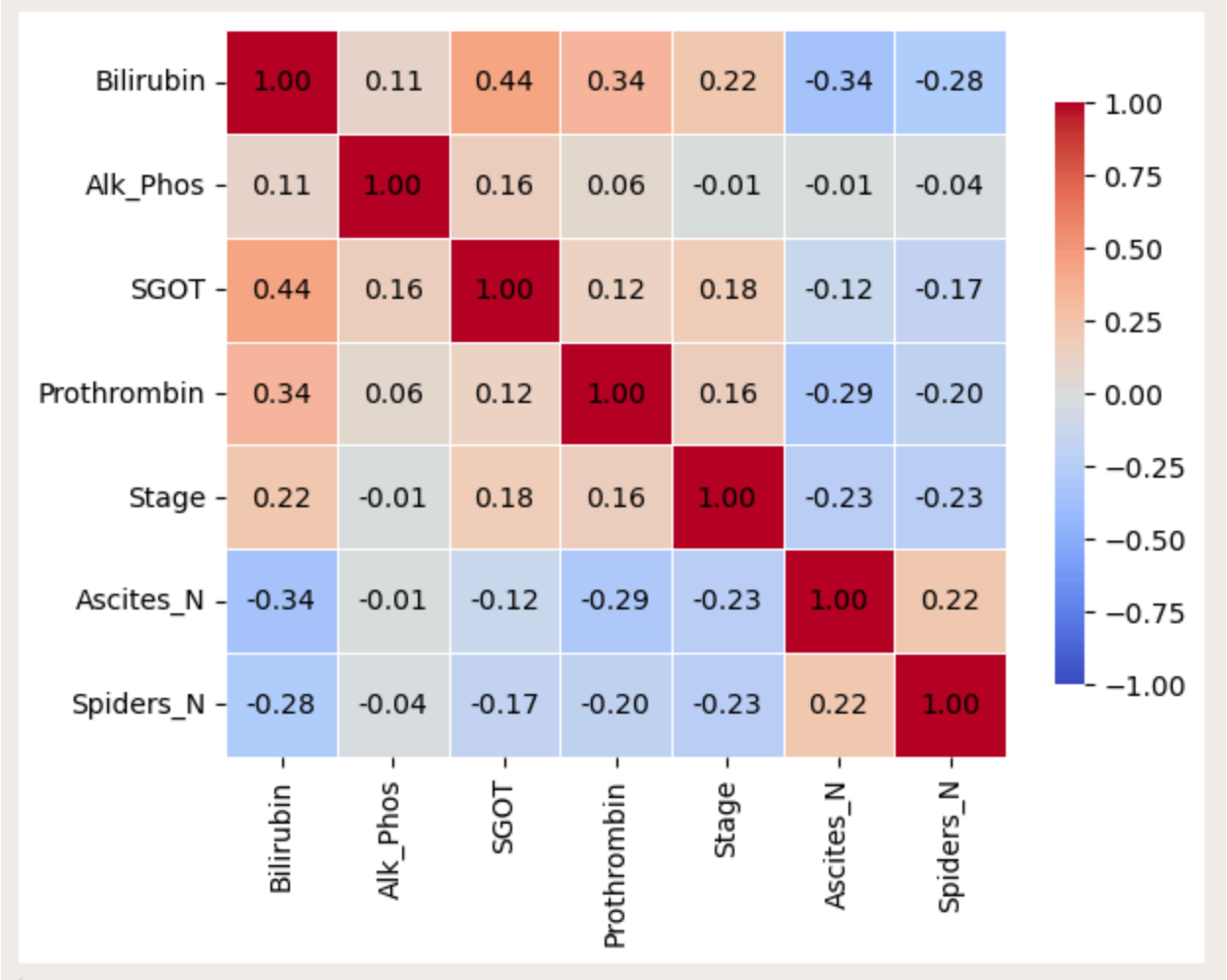
The top seven features selected by feature ranking are:
['Bilirubin', 'Alk_Phos', 'SGOT', 'Prothrombin', 'Stage', 'Ascites_N', 'Spiders_N']

Performance - Feature Ranking vs Original Feature Set

```
Feature Ranking: 0.6538
All Features: 0.6947
```

All Features performs better than Feature Ranking, 4 percentage points better. This would mean that all features included are better classifiers on an unknown datapoint than our selected 7,.

With our Pearson Correlation for the selected features, we can see that the absence of ascites and spiders are negatively correlated with all other features except each other (which can be translated to the presence of thesis positively correlated with all other selected features).



Sequential Backward Selection

(a) Setting d to seven and running the sequential selection algorithm.

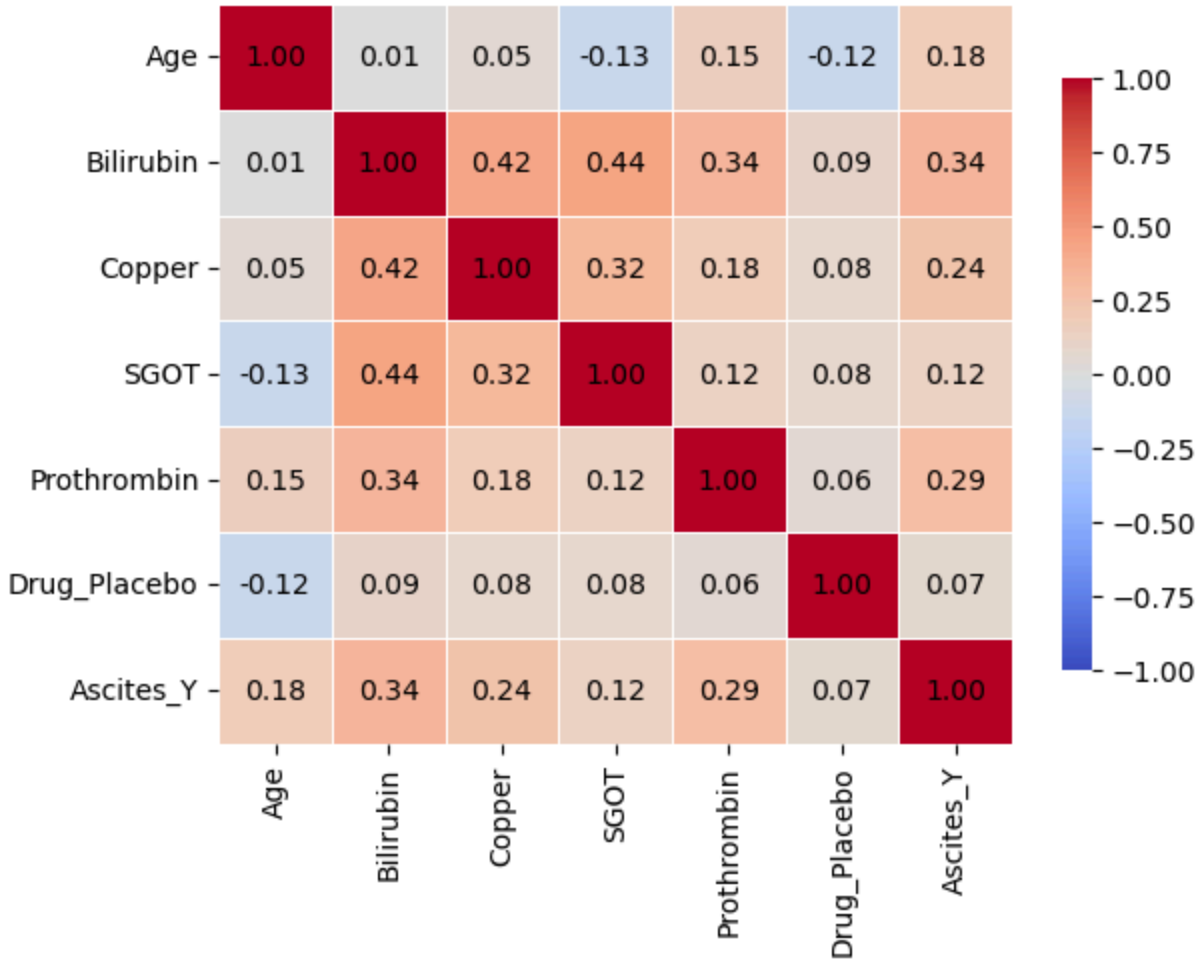
• Report the seven selected features and the test performance achieved using these seven features with KNN (K=5).

The seven selected features using SBS are:

['Age', 'Bilirubin', 'Copper', 'SGOT', 'Prothrombin', 'Drug_Placebo', 'Ascites_Y']

With a test performance of 0.6575

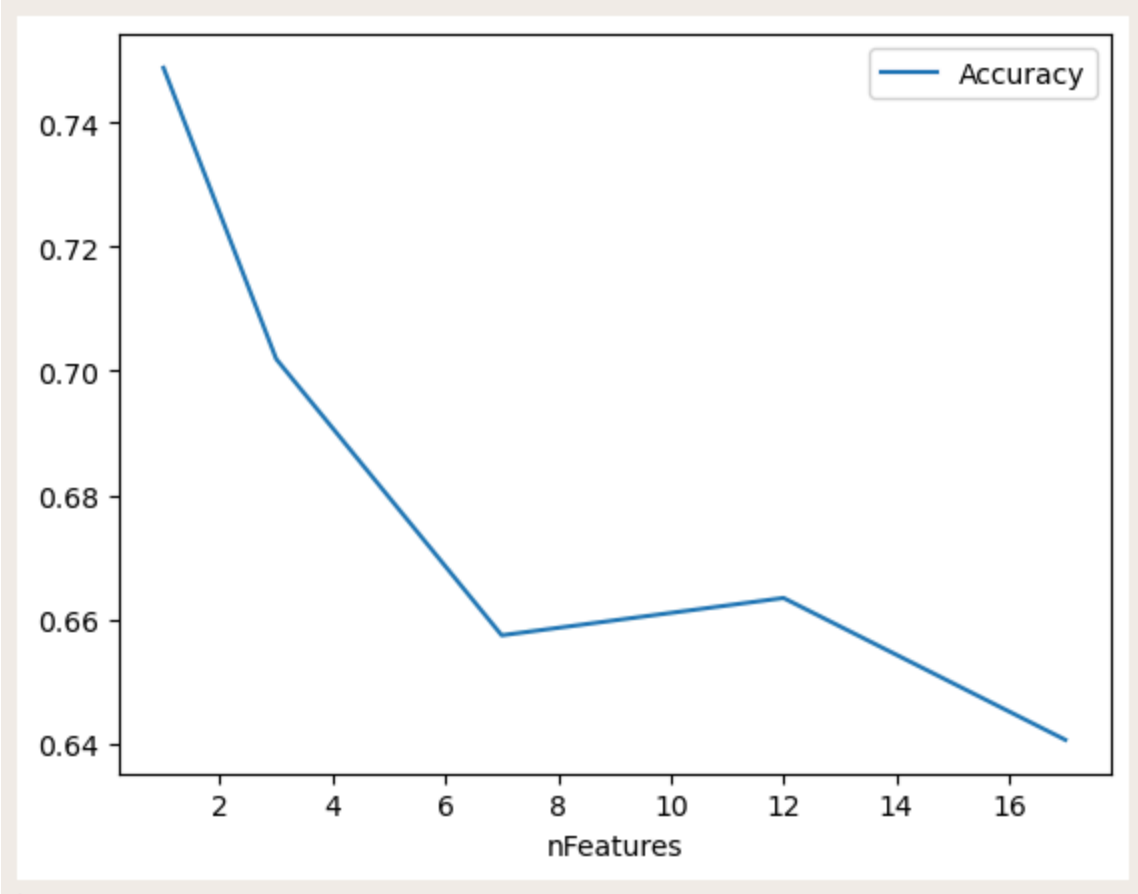
Pearson Correlation



SBS performs slightly better than feature ranking, due to how features are selected.

```
Sequential Feature Selection: 0.6575
Feature Ranking: 0.6538
```

Plotting test accuracy against no. selected features:



	nFeatures	Accuracy
0	1.0	0.748798
1	3.0	0.701923
2	7.0	0.657452
3	12.0	0.663462
4	17.0	0.640625

Our graph shows that accuracy declines the more features included in the model, with the model with the highest accuracy only having one variable - the feature with the strongest correlation with the target variable. As SBS is forced to choose more features, it starts to pick compliment features, suggesting that features not included on d=17 would be useless in the model.

```
Sequential Feature Selection (1 features): 0.7488

Index(['Bilirubin'], dtype='object')

Sequential Feature Selection (3 features): 0.7019

Index(['Bilirubin', 'Copper', 'SGOT'], dtype='object')

Sequential Feature Selection (7 features): 0.6575

Index(['Age', 'Bilirubin', 'Copper', 'SGOT', 'Prothrombin', 'Drug_Placebo',
       'Ascites_Y'],
      dtype='object')

Sequential Feature Selection (12 features): 0.6635

Index(['Age', 'Bilirubin', 'Copper', 'SGOT', 'Triglycerides', 'Platelets',
       'Prothrombin', 'Drug_D-penicillamine', 'Drug_Placebo', 'Sex_M',
       'Ascites_Y', 'Edema_S'],
      dtype='object')

Sequential Feature Selection (17 features): 0.6406

Index(['Age', 'Bilirubin', 'Cholesterol', 'Copper', 'SGOT', 'Triglycerides',
       'Platelets', 'Prothrombin', 'Drug_D-penicillamine', 'Drug_Placebo',
       'Sex_M', 'Ascites_Y', 'Hepatomegaly_N', 'Hepatomegaly_Y', 'Spiders_Y',
       'Edema_S', 'Edema_Y'],
      dtype='object')
```